

ODWA NOMBAMBELA

102 Allcock Street, Colbyn, Pretoria, 0083

071 553 0288 | 082 373 8098 | o.nombambela@gmail.com

LinkedIn: www.linkedin.com/in/odwa-nombambela | Portfolio: <https://mbizo22.github.io/Professional-Profile>

PROFESSIONAL SUMMARY

Computer Engineering graduate with hands-on experience in embedded systems, computer vision, and machine learning. Developed an automated 3D plant phenotyping system achieving 93.4% biomass estimation accuracy using dual RGB-D cameras, custom hardware integration, and advanced point cloud processing. Strong foundation in Python programming, hardware-software integration, and first-principles algorithm implementation. Seeking opportunities to apply technical skills in embedded systems, computer vision, or IoT development.

EDUCATION

Bachelor of Engineering in Computer Engineering

University of Pretoria | Expected Graduation: 2025

Relevant Coursework: Embedded Systems, Computer Vision, Machine Learning, Digital Signal Processing, Microcontrollers, Software Engineering, Data Structures & Algorithms

Short Course: I-SET Robotics Components and Pedagogy

University of South Africa (UNISA) | 2022 (Completed)

Matric Certificate

Selborne College | 2015 (Completed)

TECHNICAL PROJECTS

Automated 3D Plant Phenotyping System | Final Year Project (EPR402)

University of Pretoria | Supervisor: Prof. H. Myburgh | February 2025 - November 2025

- Designed and implemented a complete automated system for non-destructive plant biomass estimation using dual Kinect V2 RGB-D cameras mounted on a custom 6×3×2m gantry structure
- Developed five-stage processing pipeline from scratch: preprocessing, sequential ICP registration, 3D reconstruction, dimensional analysis, and Random Forest prediction
- Achieved 93.4% biomass estimation accuracy exceeding 90% target specification through voxel-based volume calculations and optimized 6-feature Random Forest model
- Implemented first-principles algorithms using NumPy for point cloud registration (RMSE: 14.1mm), grid-based mesh reconstruction, and dimensional analysis
- Integrated embedded hardware: ODROID N2+ with Arduino Uno motor controllers, NEMA 23 stepper motors with TB6600 drivers, coordinated via TCP socket communication
- Created comprehensive GUI using tkinter with multi-threaded architecture for real-time hardware control, image capture, and data visualization
- Delivered complete analysis within 2-minute target timeframe with robust error handling and graceful degradation capabilities

WORK EXPERIENCE

IT Support & Administrative Assistant

NS. Nombambela Inc | January 2022 - December 2022

- Managed technology infrastructure maintenance and provided technical IT consultation to staff
- Handled front desk operations including customer inquiries, appointment scheduling, and administrative coordination
- Maintained document management system and ensured efficient office operations and communication workflows
- Provided technical troubleshooting and support for office hardware and software systems

Community Based Project Volunteer (EBIT WEEK - JCP203)

University of Pretoria | June 2018 - July 2018

- Provided guidance to prospective students and answered questions about university programs and campus life
- Conducted campus tours for prospective students and parents, showcasing university facilities and resources
- Facilitated communication between prospective students, current students, and university representatives

Tutor - Professional Orientation (JPO110 & JPO120)

University of Pretoria | February 2017 - November 2017

- Assisted lecturers during classroom sessions, supporting student learning and engagement
- Evaluated and marked student assignments, providing detailed feedback on performance
- Mentored students through one-on-one consultations, offering academic guidance and support

Acts of Random Kindness (ARK) - Coordinator

Selborne College | 2013 - 2015

- Coordinated communications and organized meetings between school chapter members and partner institutions
- Led community service initiatives including soup kitchens and fundraising events (bake sales)
- Organized charitable donations and volunteer activities with Breath of Life NPO and Pet Pals animal shelter
- Developed leadership and project management skills through multi-year community engagement

TECHNICAL SKILLS

Programming Languages: Python (NumPy, OpenCV, tkinter, socket programming), C/C++, Java, MATLAB, Assembly, Arduino

Computer Vision & ML: 3D Reconstruction, Point Cloud Processing, ICP Registration, Random Forest, Feature Engineering, Image Processing

Embedded Systems: Arduino Uno, ODROID N2+, Kinect V2 RGB-D Cameras, Stepper Motor Control (TB6600 drivers), Serial Communication

Software Tools: Git/GitHub, Fusion 360 (CAD), LaTeX/TeXstudio, Visual Studio Code, Linux/Ubuntu

Hardware Integration: Sensor Integration, Motor Control Systems, Client-Server Architecture, TCP/IP Protocols, Robotics Components

Professional Skills: Technical Report Writing, Algorithm Development, System Design, Hardware-Software Integration, Debugging, Tutoring & Mentoring

CERTIFICATIONS & TRAINING

- CISCO CCNA Certified
- I-Set Robotics Components and Pedagogy - UNISA
- First Aid Level 3

LANGUAGES

English (Fluent) | IsiXhosa (Fluent) | IsiZulu (Intermediate) | Afrikaans (Basic)

INTERESTS & ACTIVITIES

Programming & Software Development | Photography | Sports (Squash, Rugby) | Gaming | Technical Reading

REFERENCES

Mr Andile Baleni

Associate at NS. Nombambela Inc

Cell: 082 432 6403

Mr Tony Yard

Landlord

Cell: 082 374 4421